Business Problem

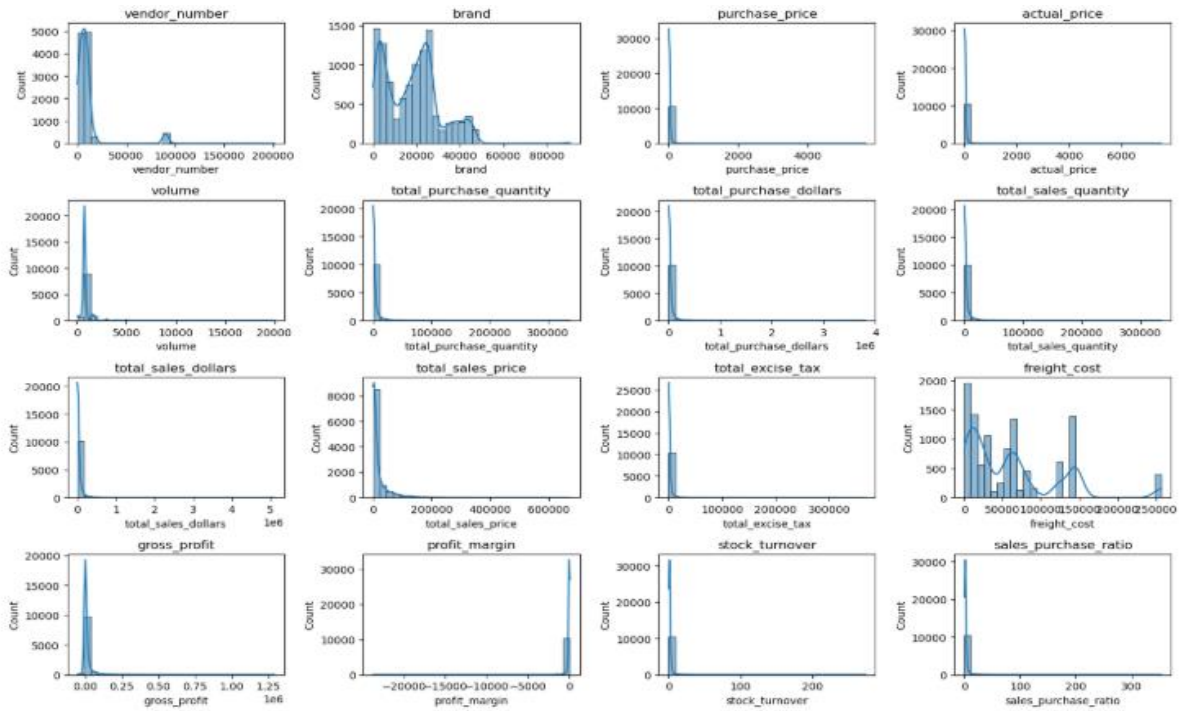
Effective Inventory and sales management are critical for optimizing profitability in the retail and wholesale industry. Companies need to ensure that they are not incurring losses due to inefficient pricing, poor inventory turnover, or vendor dependency. The goal of this analysis is to :

- Identify underperforming brands that requires promotional or pricing adjustment.
- Determine top vendors contributing to sales and gross profit.
- Impact of bulk purchasing on unit costs.

Exploratory Data Analysis Insights

Summary Statistics

	count	mean	std	min	25%	50%	75%	max
vendor_number	10692.0	1.065065e+04	18753.519148	2.00	3951.000000	7153.000000	9552.000000	2.013590e+05
brand	10692.0	1.803923e+04	12662.187074	58.00	5793.500000	18761.500000	25514.250000	9.063100e+04
purchase_price	10692.0	2.438530e+01	109.269375	0.36	6.840000	10.455000	19.482500	5.681810e+03
actual_price	10692.0	3.564367e+01	148.246016	0.49	10.990000	15.990000	28.990000	7.499990e+03
volume	10692.0	8.473605e+02	664.309212	50.00	750.000000	750.000000	750.000000	2.000000e+04
total_purchase_quantity	10692.0	3.140887e+03	11095.086769	1.00	36.000000	262.000000	1975.750000	3.376600e+05
total_purchase_dollars	10692.0	3.010669e+04	123067.799627	0.71	453.457500	3655.465000	20738.245000	3.811252e+06
total_sales_quantity	10692.0	3.077482e+03	10952.851391	0.00	33.000000	261.000000	1929.250000	3.349390e+05
total_sales_dollars	10692.0	4.223907e+04	167655.265984	0.00	729.220000	5298.045000	28396.915000	5.101920e+06
total_sales_price	10692.0	1.879378e+04	44952.773386	0.00	289.710000	2857.800000	16059.562500	6.728193e+05
total_excise_tax	10692.0	1.774226e+03	10975.582240	0.00	4.800000	46.570000	418.650000	3.682428e+05
freight_cost	10692.0	6.143376e+04	60938.458032	0.09	14069.870000	50293.620000	79528.990000	2.570321e+05
gross_profit	10692.0	1.213238e+04	46224.337964	-52002.78	52.920000	1399.640000	8660.200000	1.290668e+06
profit_margin	10692.0	-inf	NaN	-inf	13.324515	30.405457	39.956135	9.971666e+01
stock_turnover	10692.0	1.706793e+00	6.020460	0.00	0.807229	0.981529	1.039342	2.745000e+02
sales_purchase_ratio	10692.0	2.504390e+00	8.459067	0.00	1.153729	1.436894	1.665449	3.529286e+02



Negative & Zero Values:

- Gross profit: Minimum value is less than zero indicating losses. Some products or transactions may be selling at a loss due to high costs or selling at discounts lower than the purchase price.
- Profit Margin: Has a minimum of negative infinity value, which suggests cases where revenue is zero or even lower than costs.
- Total Sales Quantity & Sales Dollars: minimum values are 0, meaning some products were purchased but never sold. These could be slow-moving or obsolete stock.

Outliers Indicated by High Standard Deviations:

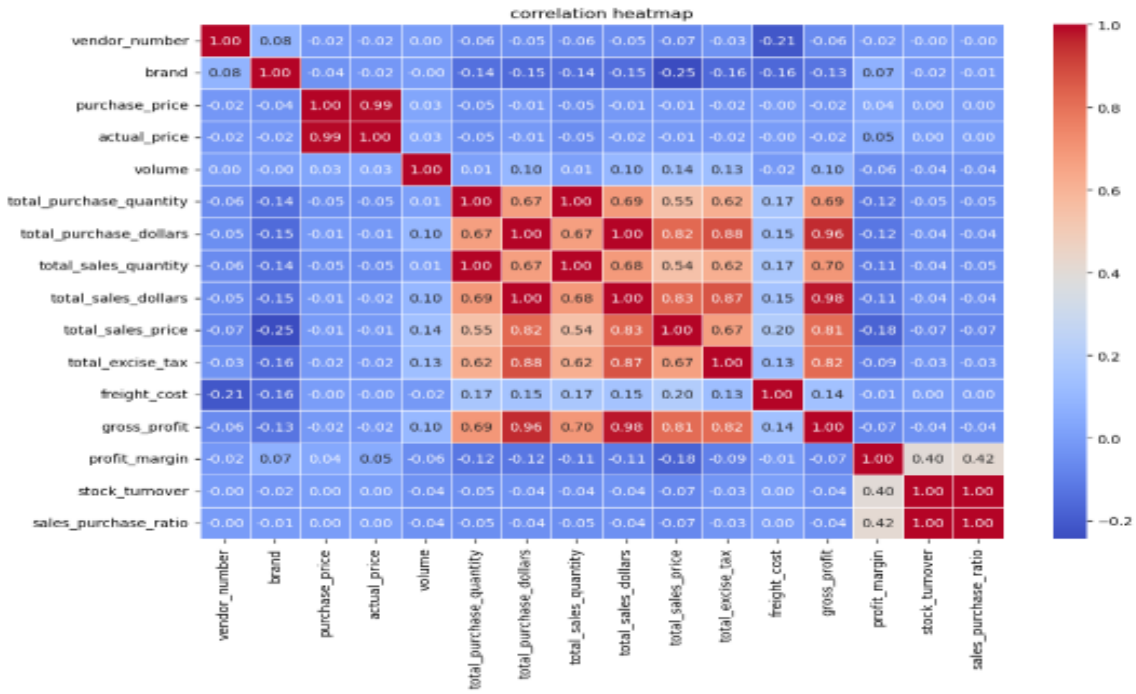
- Purchase & Actual Prices: The max values are significantly higher than the mean, indicating potential premium products.
- Freight cost: Huge variation suggests logistics inefficiencies or bulk shipments.
- Stock Turnover: Ranges from 0 to 274.5, implying some products sell extremely fast while others remain in stock definitely. Value more than 1 indicates that sold quantity for that products is higher than purchased quantity due to either sales are being fulfilled from older stock.

Data Filtering

To enhance the reliability of the insights, we removed inconsistent data points where:

- Gross profit <= 0 (to exclude transactions leading to losses.)
- Profit margin <= 0 (to ensure analysis focuses on profitable transactions.)
- Total sales quantity = 0 (to eliminate inventory that was never sold)

Correlation Insights



- Purchase price has weak correlation with total sales dollars (-0.01) and gross profit (-0.02), suggesting that price variations do not significantly impact sales revenue or profit
- Strong correlation between total purchase quantity and total sales quantity, confirming efficient inventory turnover.
- Negative correlation between profit margin and total sales price suggests that as sales price increases, margins decrease, possibly due to competitive pricing pressures.
- Stock turnover has weak negative correlations with both gross profit and profit margin, indicating that faster turnover does not necessarily result in higher profitability

Research Questions & Key findings

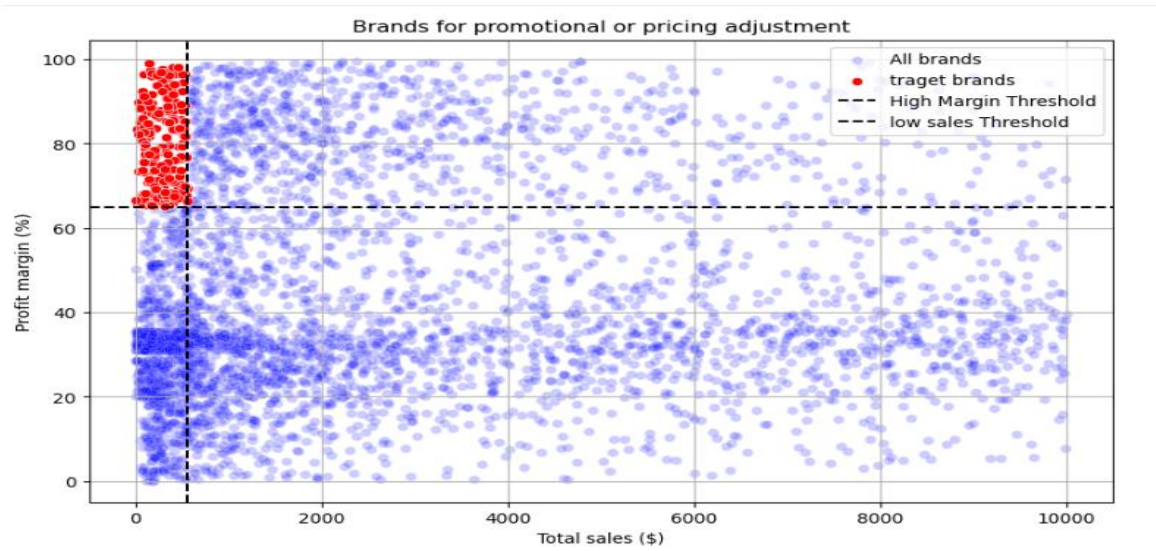
1.Identify Brands that needs promotional or pricing adjustments which exhibit lower sales performance but higher profit margins.

```
brands with low sales but high profit margin:
```

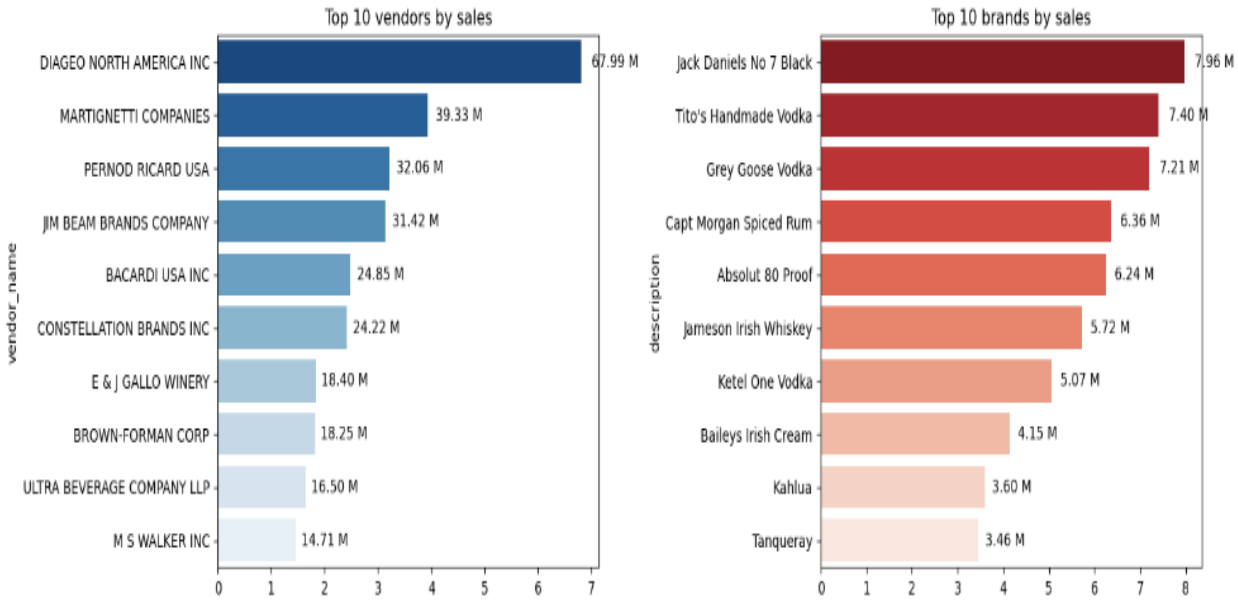
	description	total_sales_dollars	profit_margin
6199	Santa Rita Organic Svgn Bl	9.99	66.466466
2369	Debauchery Pnt Nr	11.58	65.975820
2070	Concannon Glen Ellen Wh Zin	15.95	83.448276
2188	Crown Royal Apple	27.86	89.806174
6237	Sauza Sprklg Wild Berry Marg	27.96	82.153076
...	...	...	...
5074	Nanbu Bijin Southern Beauty	535.68	76.747312
2271	Dad's Hat Rye Whiskey	538.89	81.851584
57	A Bichot Clos Marechaudes	539.94	67.740860
6245	Sbragia Home Ranch Merlot	549.75	66.444748
3326	Goulee Cos d'Estournel 10	558.87	69.434752

198 rows × 3 columns

198 brands exhibit lower sales but higher profit margins, which could benefit from targeted marketing, promotions, or price optimizations to increase volume without compromising profitability.

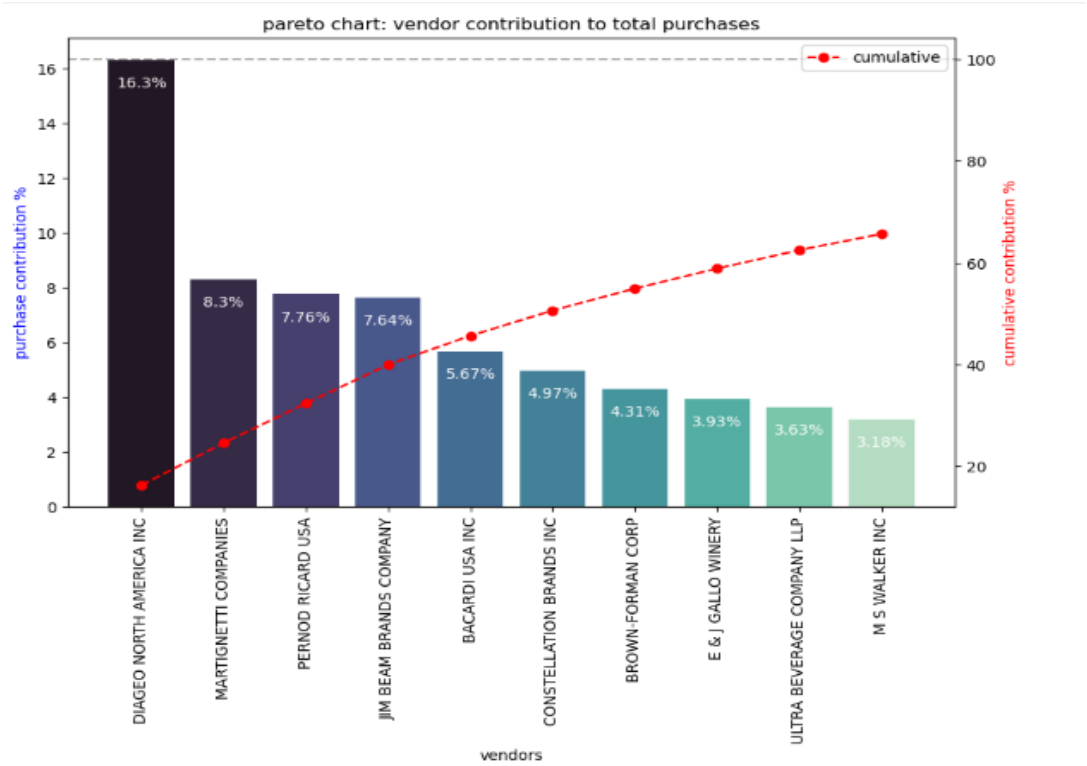


2.Which vendors and brands demonstrate the highest sales performance?



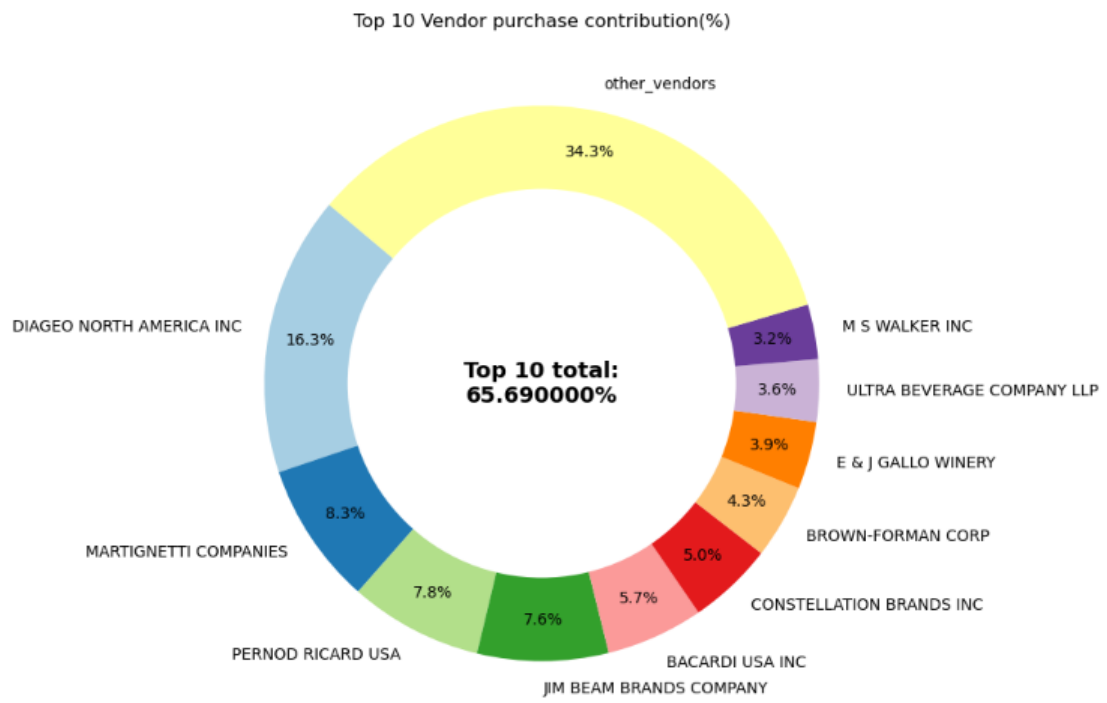
Vendors such as **Diageo North America Inc.** dominate vendor sales, while brands like **Jack Daniels No. 7 Black**, **Tito's Handmade Vodka**, and **Grey Goose Vodka** lead in brand-level sales. This indicates that a few key vendors and brands drive a major portion of revenue, highlighting their strong market demand and importance in overall sales growth.

3.Which vendors contribute the most to total purchase dollars?



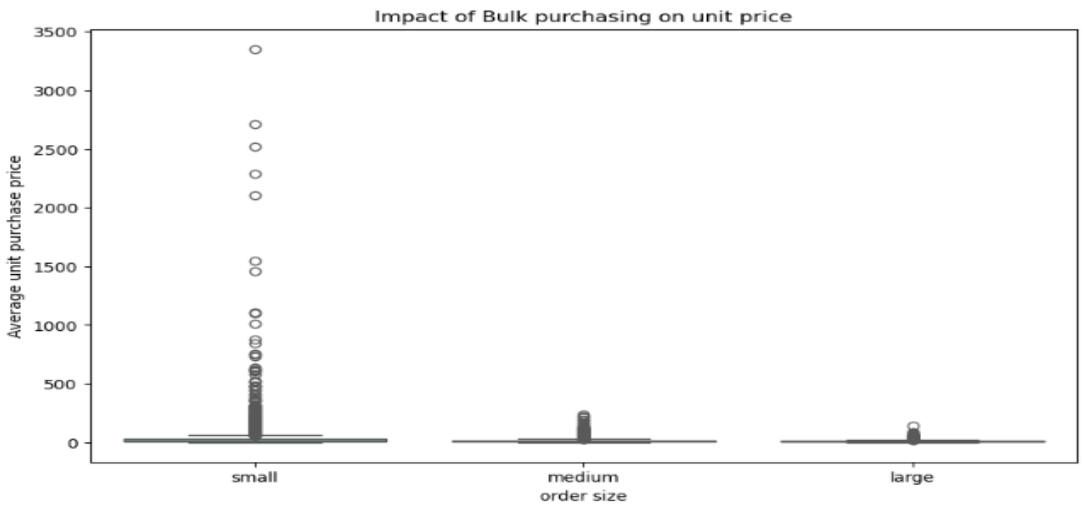
Diageo North America Inc. contributes the highest share at 16.3%, followed by Martignetti Companies (8.3%) and Pernod Ricard USA (7.76%). The cumulative line indicates that a small group of top vendors accounts for a significant portion of total purchases.

4.How much of total procurement is dependent on the top vendors?



Top 10 Vendors contribute 65.69% of total purchase, while the remaining vendors contribute only 34.31%. This over-reliance on a few vendors may introduce risks such as supply chain disruptions, indicating a need for diversification.

5.Does bulk purchasing reduce the unit price and what is the optimal purchase volume for cost savings?



- Vendors buying in bulk (Large order size) get the lowest unit price, meaning higher margins if they can manage inventory efficiently.
- The price difference between small and large order is substantial (~72% reduction in unit cost)

- This suggests that bulk pricing strategies successfully encourage vendors to purchase in large volumes, leading to higher overall sales despite lower per-unit revenue.

Recommendations

- Diversify vendor partnership to reduce dependency on a few suppliers and mitigate supply chain risks
- Leverage bulk purchasing advantages to maintain competitive pricing while optimization inventory management.
- Enhance marketing and distribution strategies for low-performing vendors to drive higher sales volumes without compromising profit margins.
- By implementing these recommendations, the company can achieve sustainable profitability, mitigate risks, and enhance overall operational efficiency.